

CSE422 Lab Project Report Template

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1. Introduction

- Select one feature with discrete values as your target feature. [Note: You may discretize one feature with continuous values as well.]
- Give a small introduction to the project and its aim. Describe the problem the project is aiming to solve and the motivation behind the project.

2. Dataset Description

Description should contain the following:

- Classification or regression problem? Why do you think so?
- Dataset size and Number of features
- Types of features in the dataset (Quantitative / Categorical)
- Dataset Balance / Imbalance
 - For the output feature, do all unique classes have an equal number of instances or not?
 - Represent using a bar chart of N classes (N=number of classes you have in your dataset).

3. Data Analysis

- Perform exploratory data analysis to extract some important relationships from your data. [Reference:  EDA Lab CSE422]
- What do you understand after the correlation test?

4. Dataset Pre-processing

- Handling null and duplicate values
- Encoding categorical values
- Feature Selection
- Feature scaling
- Other feature engineering, if necessary

Note: Discuss one problem, and then write about the solutions or pre-processing techniques you have applied to solve that problem

5. Dataset Splitting

- Random/Stratified (as required)
- Train set (80% / 70%) (Use Validation Set as required)
- Test set (20% / 30%)

6. Model Training & Testing (Supervised)

- [KNN](#) (for classification problem)
- [Decision Tree](#) (for classification/regression problem)
- [Logistic Regression](#) (for classification problem)
- [Linear Regression](#) (for regression problem)
- [Naive Bayes](#) (for classification problem)
- [Neural Network](#) (for classification/regression problem) [This can be applied using any library you feel comfortable with - sklearn, tensorflow, pytorch etc]

You have to apply a Neural Network and at least 2 other models (Total 3 at least, but you are encouraged to do all)

7. Model selection/Comparison analysis

- Bar chart showcasing prediction accuracy of all models (for classification)
- Precision, recall, F1 score comparison of each model. (for classification)
- Confusion Matrix (for classification)
- AUC score, ROC curve for each model (for classification)
- R^2 score and Loss (for regression)

Compare the results of all models based on all of the above described metrics

8. Unsupervised Learning

- **Remove the target / label from the dataset and treat the problem as an unsupervised learning problem**
- **Apply K-Means Clustering and show the clusters**

9. Conclusion

- Best model based on the evaluations in 7
- If you are getting low performance metrics, what can be potential reasons?
- What are some of the challenges that you have faced?

Report Template [Overleaf, Preferred]:

<https://www.overleaf.com/latex/templates/labreport-ieee-journal-paper-template/ypbfinkbvshw>

[You can use Google Docs / MS-Word if you cannot use Overleaf]